

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	C	A	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,C	A,C	B,C,D	B,C,D	A,B,D	A,B,D	B,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	3.11	8.00	9.00	9.00	37.00	10.00		

PART-2 : CHEMISTRY

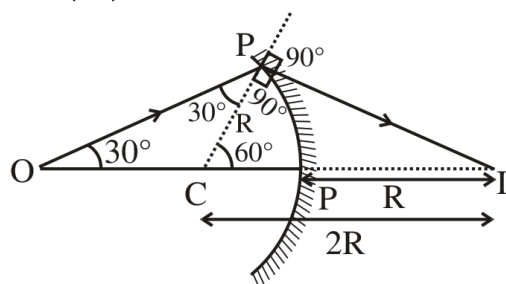
SECTION-I (i)	Q.	1	2	3	4				
	A.	D	A	A	C				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	B,C	A,D	B,D	A,B,D	B,C	A,B,C	A,B,D	C,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	9.00	18.00	7.00	3.00	8.00	5.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	D	A	A	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	B,D	C,D	A,B,C	A,B,D	B,C,D	A,D	A,B,D	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	6.00	2.00	8.50	3.00	442.50	285.00		

HINT – SHEET**PART-1 : PHYSICS****SECTION-I (i)**

1. Ans (A)



By snell's law

$$2\sqrt{3} \sin 30 = \sqrt{3} \sin(r)$$

$$\sin r = 1 \Rightarrow r = 90^\circ$$

In ΔCPI

$$\frac{CP}{CI} = \cos 60^\circ$$

$$\Rightarrow CI = 2R$$

3. Ans (A)

Let refractive index of glass be μ Let after first refraction, image distance be v then

$$\frac{\mu}{v} - \frac{1}{\infty} = \frac{\mu - 1}{R} \Rightarrow v = \frac{\mu R}{\mu - 1}$$

Now second refraction will take place. So distance of

$$\text{first image from O is } u_1 = \frac{\mu R}{\mu - 1} - R = \frac{R}{\mu - 1}$$

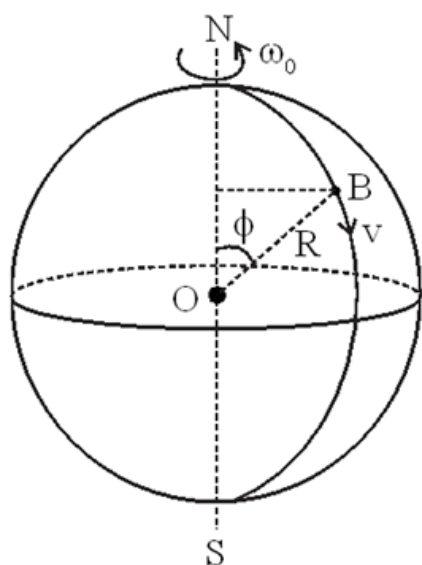
And image is formed at R

$$\therefore \frac{1}{R} - \frac{\mu(\mu - 1)}{R} = \frac{2(1 - \mu)}{R} \Rightarrow \mu^2 - 3\mu + 1 = 0$$

$$\text{So } \mu = \frac{3 + \sqrt{5}}{2}$$

4. Ans (A)

The bug travels a distance πR as it move from the north pole to the south pole with a speed v (relative to the spherical globe). The time taken by the bug to complete this journey is $T = \pi R/v$.



$I = \frac{2}{5}MR^2 + mR^2\sin^2\phi$. The component of angular momentum of the system along SN at this instant is $L_z = \left(\frac{2}{5}MR^2 + mR^2\sin^2\phi\right)\omega$.

The conservation of angular momentum,

$L_{z0} = L_z$, gives angular speed of the sphere at

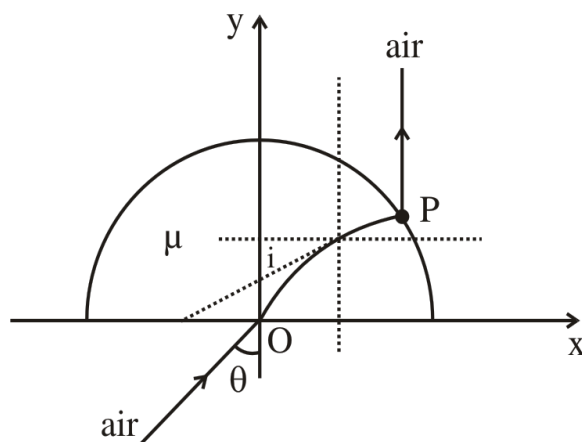
time t

$$\omega = \frac{I_0\omega_0}{I} = \frac{2M\omega_0}{2M + 5m\sin^2\phi} = \frac{2M\omega_0}{2M + 5m\sin^2(vt/R)}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,C)



$$\frac{dy}{dx} = \tan i$$

Now using Snells law

$$1 \times \sin 90^\circ = \mu \sin i$$

$$\sin i = \frac{1}{\mu} = \left(\frac{a-x}{a}\right)$$

$$\sin i = \left(\frac{a-x}{a}\right)$$

$$\tan i = \frac{(a-x)}{\sqrt{a^2 - (a-x)^2}} \quad \dots(i)$$

$$\frac{dy}{dx} = \frac{(a-x)}{\sqrt{2ax - x^2}} \text{ from equation (i)}$$

$$\int_0^y dy = \int_0^x \frac{(a-x)}{\sqrt{2ax - x^2}} dx \Rightarrow y = \sqrt{2ax - x^2} \quad \dots(ii)$$

$$\text{Also, } y^2 + x^2 = \frac{a^2}{4} \quad \dots(iii)$$

Solving (ii) and (iii) we get

$$2ax = \frac{a^2}{4} \Rightarrow x = \frac{a}{8} \text{ and } y = \frac{a}{8}\sqrt{15}$$

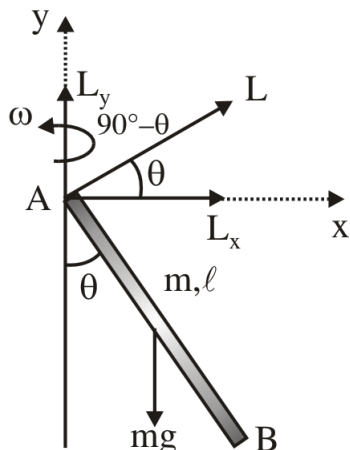
The x and y coordinates of point P are

$$\left(\frac{a}{8}, \frac{a}{8}\sqrt{15}\right)$$

6. Ans (A,C)

Total angular momentum of the rod about end

$$A \text{ is } L = \frac{m\ell^2}{3} \omega \sin \theta$$



$$L_x = L \cos \theta = \frac{m\ell^2}{3} \omega \sin \theta \cos \theta$$

$$L_y = L \sin \theta = \frac{m\ell^2}{3} \omega \sin^2 \theta$$

$$\vec{L} = \vec{L}_x + \vec{L}_y$$

$$\frac{d\vec{L}}{dt} = \frac{d\vec{L}_x}{dt} + \frac{d\vec{L}_y}{dt}$$

$$\frac{d\vec{L}}{dt} = \frac{d\vec{L}_x}{dt}$$

$$\vec{\tau} = \frac{d\vec{L}}{dt} = \frac{d\vec{L}_x}{dt}$$

$$mg \frac{\ell}{2} \sin \theta = L_x \omega$$

$$mg \frac{\ell}{2} \sin \theta = \frac{m\ell^2}{3} \omega^2 \sin \theta \cos \theta$$

$$\omega = \sqrt{\frac{3g}{2\ell \cos \theta}} = \sqrt{\frac{3 \times 10 \times 5}{2 \times 1 \times 3}} = 5 \text{ rad/s}$$

Now let F_h and F_v are the horizontal and vertical components of hinge reaction acting on the rod at end A

$$F_h = \frac{m\omega^2 \ell}{2} \sin \theta = 1 \times 25 \times \frac{1}{2} \times \frac{4}{5} = 10 \text{ N}$$

$$F_v = mg = 10 \text{ N}$$

The net hinge reaction

$$F = \sqrt{F_h^2 + F_v^2} = 10\sqrt{2} \text{ N}$$

7. Ans (B,C,D)

Let the acceleration of block is a downward and the tension in the string connected with the cylinder is T.

For the block

$$64mg - 16T = 64ma \quad \dots(i)$$

For the cylinder

$$T + f = 8ma \quad \dots(ii)$$

Torque about contact point

$$T \times 2R = \frac{3}{2} mR^2 \alpha$$

$$R\alpha = 8a \quad \dots(iii)$$

$$T = 6ma \quad \dots(iv)$$

Solving (i), (ii), (iii) and (iv)

$$T = 48N, a = 4 \text{ m/s}^2 \text{ and } f = 16N$$

8. Ans (B,C,D)

From conservation of angular momentum

$$(2mx^2 + I)\omega' = (2mR^2 + I)\omega$$

$$\Rightarrow \omega' = \frac{(2mR^2 - I)\omega}{2mx^2 + I}$$

Force exerted by insect on the turntable

$$F = m\omega'^2 x = m \frac{(2mR^2 - I)^2 \omega^2}{(2mx^2 + I)^2} x$$

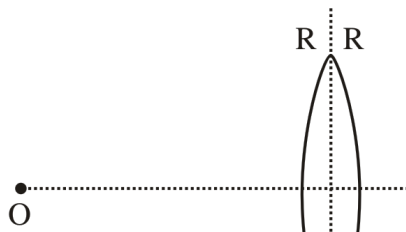
$$\text{For } F \text{ to maximum } \frac{dF}{dx} = 0$$

$$\Rightarrow x = \sqrt{\frac{1}{6m}} = \sqrt{\frac{\frac{2}{3}mR^2}{6m}} = \frac{R}{3}$$

$$\omega' = \left(\frac{2mR^2 + \frac{2}{3}mR^2}{2m\frac{R^2}{9} + \frac{2}{3}mR^2} \right) \omega = 3\omega$$

$$F_{\max} = m\omega'^2 x = m \times 9\omega^2 \times R/3 = 3m\omega^2 R$$

9. Ans (A,B,D)



$$\frac{1}{f} = \left(\frac{3}{2} - 1\right) \left(\frac{1}{R} + \frac{1}{R}\right) = \frac{1}{R} \Rightarrow f = R$$

$$\frac{1}{f_w} = \left[\frac{3/2}{4/3} - 1\right] \left(\frac{1}{R} + \frac{1}{R}\right) = \frac{1}{4f} \Rightarrow f_w = 4f$$

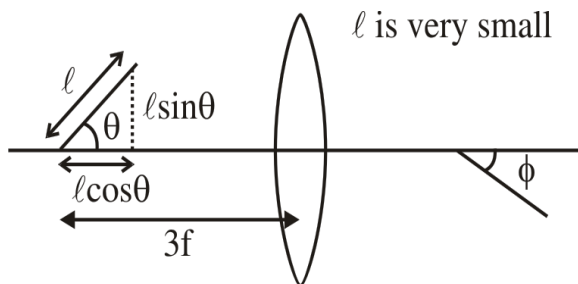
When in water:

$$\frac{1}{4f} = \frac{1}{v'} + \frac{1}{d} \Rightarrow \frac{1}{v'} = \frac{1}{4f} - \frac{1}{d}$$

$$\Rightarrow v' = \frac{4fd}{d - 4f} < 0 \text{ and } \frac{v'}{u} > 1$$

$\therefore$ Image is virtual, magnified and erect.

10. Ans (A,B,D)



$$m = \frac{f}{f + u} = \frac{-1}{2}$$

$$dv = m^2 du = \frac{l \cos \theta}{4}$$

$$n_i = \text{hom} = -\frac{l \sin \theta}{2}$$

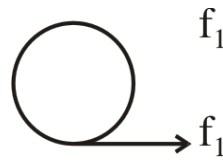
$$\tan \theta = \frac{n_i}{dv} = \frac{l \sin \theta}{2} \times \frac{4}{l \cos \theta} = 2 \tan \theta$$

$$\tan \theta = 2 \tan 30^\circ = \frac{2}{\sqrt{3}}$$

$$\sec^2 \theta \frac{d\theta}{dt} = 2 \sec^2 \theta \frac{d\theta}{dt}$$

$$\frac{d\phi}{dt} = \frac{2 \cos^2 \phi}{\cos^2 \theta} \frac{d\theta}{dt} = 2 \times \left(\frac{\sqrt{3}}{\sqrt{7}}\right) \times \frac{1}{\left(\frac{\sqrt{3}}{2}\right)^2} = 7$$

11. Ans (B,D)



$$f_1 + f_2 = m a_{cm} \quad \text{.....(i)}$$

$$f_1 R - f_2 R = I_{cm} \alpha \quad \text{.....(ii)}$$

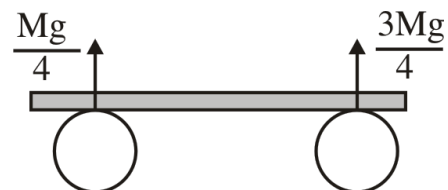
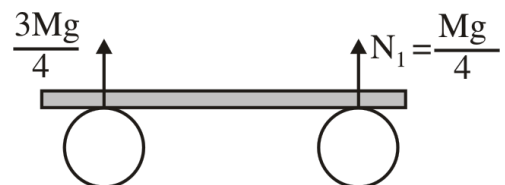
$$a_{cm} = R \alpha \quad \text{.....(iii)}$$



$$F - 2f_1 = M(2a_{cm}) \quad \text{.....(iv)}$$

$$a_{cm} = \frac{F}{2(M + m)}$$

$$V = \sqrt{2(2a_{cm}) \frac{2L}{3}}$$



$$f_1 = \frac{mF}{\mu(M + m)} = \mu \left(\frac{Mg}{4}\right)$$

$$\mu = \frac{4mF}{2M(M + m)g} = \frac{2mF}{M(M + m)g}$$

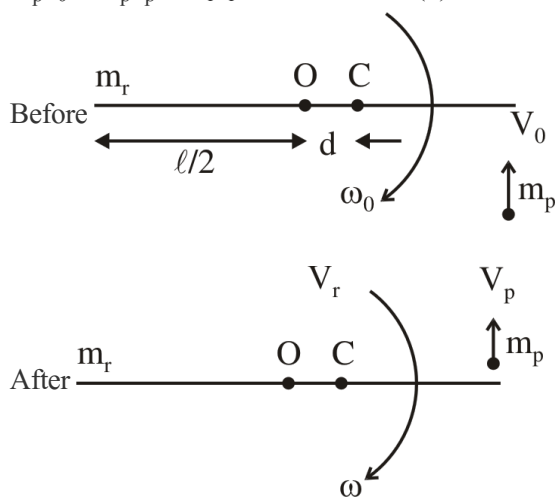
12. Ans (A,B,C)

Consider the rod and the particle together as a system. There is no external force on the system. Thus, linear momentum as well as angular momentum of the system are conserved.

Let w be angular velocity of the rod, v_r be velocity of its centre of mass and v_p be velocity of the particle immediately after the impact.

The conservation of linear momentum gives

$$m_p v_0 = m_p v_p + m_r v_r \quad \dots\dots(1)$$



The center of mass C of the system lies at a distance $d = 1/6$ from the rod's centre O.

Before collision, angular momentum of the system about a fixed point on the table that coincides with C (at the time of collision) is

$$\begin{aligned} L_0 &= m_p v_0 \left(\frac{1}{2} - d \right) - I_C \omega_0 \\ &= m_p v_0 \left(\frac{1}{2} - \frac{1}{6} \right) - \left(\frac{1}{12} m_r l^2 + m_r \left(\frac{1}{6} \right)^2 \right) \omega_0 \\ &= \frac{2}{3} m_p v_0 - \frac{1}{9} m_r l^2 \omega_0. \end{aligned}$$

After collision, angular momentum of the system about the same point is

$$\begin{aligned} L &= \frac{2}{3} m_p v_p - \frac{1}{9} m_r l^2 \omega \\ L &= \frac{2}{3} m_p v_p - \frac{1}{9} m_r l^2 \omega. \end{aligned}$$

We assumed that the rod keeps on rotating in the same direction. The conservation of angular momentum, $L_0 = L$, gives

$$6m_p v_0 - m_r l^2 \omega_0 = 6m_p v_p - m_r l^2 \omega \quad \dots\dots(2)$$

Before collision, the velocity of approach (i.e., velocity of the particle relative to the velocity of the rod's impact point) is $v_a = v_0 + w_0 l/2$.

After collision the velocity of separation is

$$v_s = (v_r - w l/2) - v_p.$$

The coefficient of restitution e is the ratio of the velocity of separation to the velocity of approach i.e.,

$$e = \frac{v_s}{v_a} = \frac{v_r - w l/2 - v_p}{v_0 + w_0 l/2} = \frac{2}{3} \quad \dots\dots(3)$$

Substitute values and solve equations (1) – (3)

to get $v_p = 1$ m/s, $v_r = 2.5$ m/s and $w = -9$ rad/s (negative sign indicates anticlockwise rotation of the rod).

PART-1 : PHYSICS

SECTION-II

1. Ans (3.11)

For image formed by the refraction at water surface

$$\frac{\frac{4}{3}}{v_1} - \frac{1}{-48} = \frac{\frac{4}{3} - 1}{\infty}$$

$$\Rightarrow v_1 = -64 \text{ cm}$$

For the lens

$$u_2 = -(64 + 11) = -75 \text{ cm}$$

focal length of the lens in water f_w can be found by

$$\frac{f_w}{f_{\text{air}}} = \frac{(\mu - 1)}{\left(\frac{\mu}{\mu_w} - 1 \right)} = \frac{1.5 - 1}{\left(\frac{1.5}{4/3} - 1 \right)}$$

2. Ans (8.00)

For the lower half with the help of the double refraction v final co-ordinates (140cm, 0) for the upper half with the help of lens maker's formed the co-ordinate of the image formed by upper

half- of the lens is $\left(\frac{320}{3} \text{ cm}, \frac{8}{3} \text{ mm} \right)$.

Hence then difference in Y-co-ordinate of images is $8/3$ mm.

3. Ans (9.00)

For the final image to coincide with the object, the light rays are to be retracted. The silvered lens behaves like a concave mirror with focal length 12cm.

4. Ans (9.00)

Applying Snell's law at C

$$1 \sin i = \mu \sin r$$

$$\text{or } 1 \sin 60^\circ = \sqrt{3} \sin r$$

$$\text{or } \frac{\sqrt{3}}{2} = \sqrt{3} \sin r$$

$$\therefore r = 30^\circ$$

$$AC = 3 \tan 30^\circ = \sqrt{3}$$

$$\therefore \text{BD} = 4\sqrt{3} \quad \therefore \text{DE} = 4\sqrt{3} \cot 60^\circ = 4$$

$\therefore$ Coordinate of points E = (9, 0).

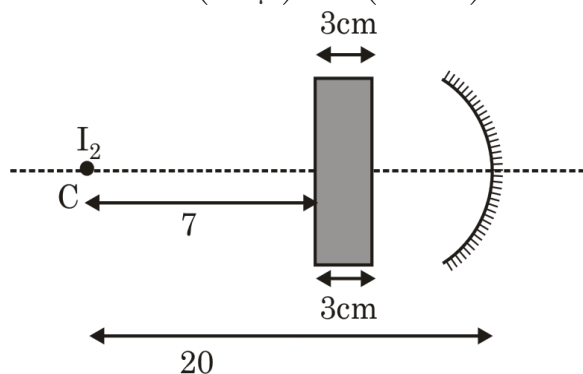
6. Ans (10.00)

For auto-collimation; image has to be formed on 'C' of mirror.

$$\frac{1}{f} = (1.5 - 1) \left(\frac{1}{-60} + \frac{1}{20} \right) = \frac{1}{60}$$

$$\text{E-1: } \frac{1}{v_1} - \frac{1}{-80} = \frac{1}{60} \Rightarrow v_1 = 240$$

$$\text{E-2 : Shift} = t \left(1 - \frac{1}{\mu} \right) = 3 \left(1 - \frac{1}{1.5} \right) = 1 \text{ cm}$$



$$\therefore x = 10$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (D)

Ester is sensitive to acid as well as base.

iv and v can reduce aldehyde to alkane.

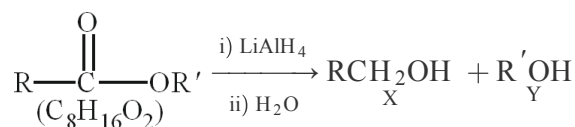
ii and iii don't reduce ester.

2. Ans (A)

(i) Ortho effect

(ii) +I, $\text{CH}_3 < \text{CD}_3 < \text{CT}_3$
$$\text{(iii) } -\text{I}, -\overset{+}{\text{N}}\text{Me}_3 > -\overset{+}{\text{N}}\text{H}_3$$

3. Ans (A)



R' must be 3°

RCH_2OH is always a 1° Alcohol.

PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (B,C)

Semipinacolone rearrangement.

When -NH_2 group is at equatorial position, ring contraction occurs due to migration of C-C bond anti to -NH_2 group.

6. Ans (A,D)

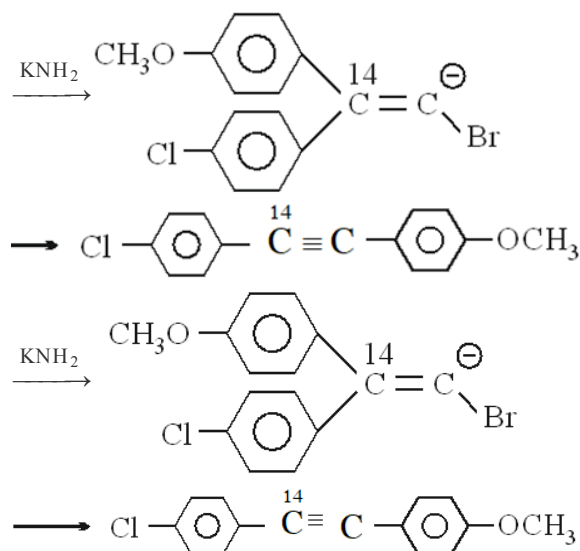
The halide used in the synthesis of grignard reagent should not contain active H or electrophilic center.

7. Ans (B,D)

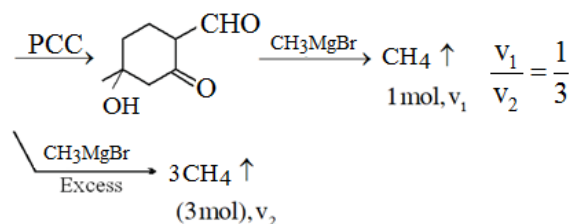
(B) H_2/pd , reduces yne to ane.

(D) Both the Br atoms will add to same C atom.

8. Ans (A,B,D)



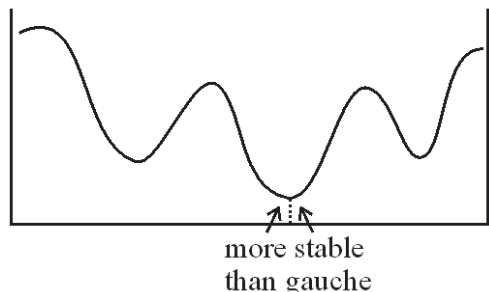
9. Ans (B,C)



11. Ans (A,B,D)

(A) Gauche forms of butane are chiral and mirror images of each other.

(B) The second most stable conformation of butane lies just in the vicinity of the anti-conformation.



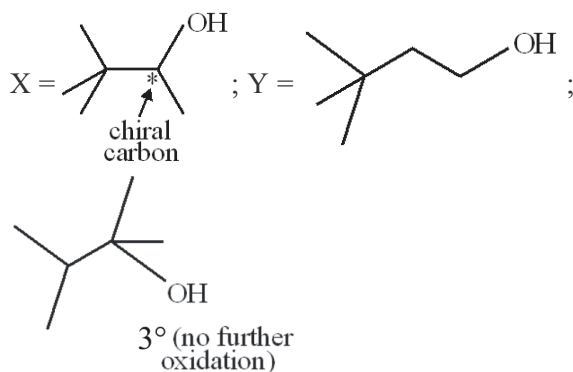
(C) Anti butane Gauche butane $\Delta H^\circ = +ve$

$\therefore \Delta H^\circ = +ve$, Relative percentage of product

increases as temperature is increased.

D) Other than anti, other conformers of  are polar. Hence the compound has a dipole moment.

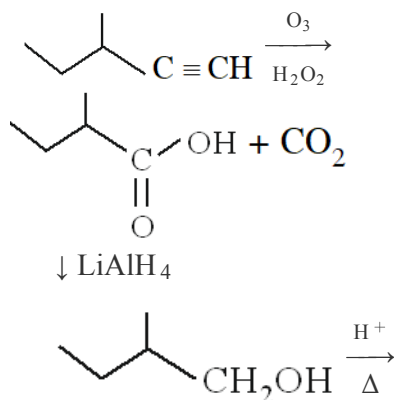
12. Ans (C,D)



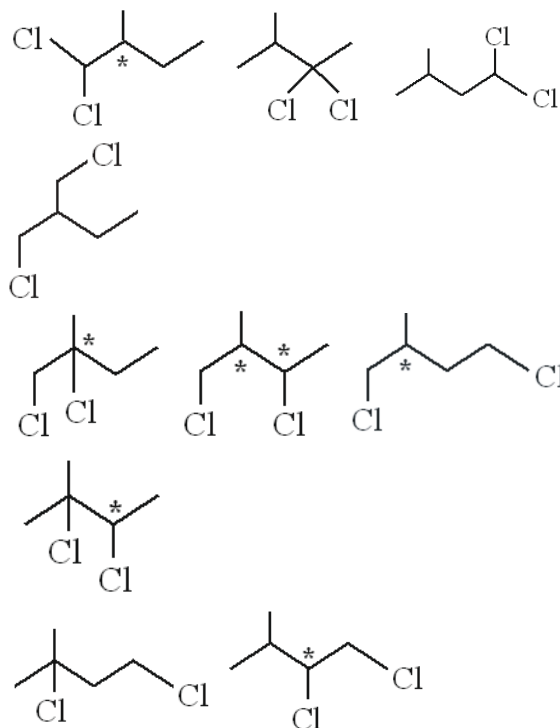
PART-2 : CHEMISTRY

SECTION-II

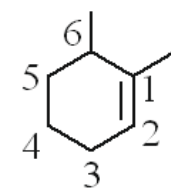
1. Ans (9.00)



2. Ans (18.00)



3. Ans (7.00)



$$\Rightarrow 1 + 6 = 7$$

4. Ans (3.00)

Racemization is the reaction in which an optically active compound is converted into an equilibrium equimolar mixture of the compound and its enantiomer.

Only reaction 1,4 and 8 undergo racemization

5. Ans (8.00)

One chiral center $2^3 = 8$

One alkene

Biphenyl system

6. Ans (5.00)

i, ii, v, vi, viii

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (D)**

$$\text{Projection} = (8\vec{a} - 3\vec{b}) \cdot \hat{c} = 8\vec{a} \cdot \hat{c} - 3\vec{b} \cdot \hat{c}$$

$$= \frac{2}{|\vec{c}|} = 2\sqrt{\frac{12}{7}}$$

$$[\vec{a} \ \vec{b} \ \vec{c}]^2 = \begin{vmatrix} 9 & 3 & 1 \\ 3 & 9 & 2 \\ 1 & 2 & c^2 \end{vmatrix} = 9$$

$$c^2 = \frac{42}{72} = \frac{7}{12}$$

2. **Ans (A)**

$$2^m = 2^n + 112$$

$$2^m - 2^n = 112$$

$$2^n(2^{m-n} - 1) = 112$$

$$2^n(2^{m-n} - 1) = 2^4(2^3 - 1)$$

$$n = 4, m = 7$$

3. **Ans (A)**

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 2), (3, 3), (3, 4), (4, 3), (4, 4), (5, 5)\}$$

4. **Ans (A)**

The domain of $f(x) = h(x) + g(x)$

The domain of $f(x)$ is $D_1 \cap D_2$

$$3 - 2^x - 2^{1-x} \geq 0$$

$$3 - 2^x - \frac{2}{2^x} \geq 0$$

$$(2^x)^2 - 3 \cdot 2^x + 2 \leq 0$$

$$(2^x - 2)(2^x - 1) \leq 0$$

$$2^x \in [1, 2]$$

$$x \in [0, 1] \quad \dots\dots D_1$$

$$\text{and domain } g \quad g(x) = \sqrt{1 - \sqrt{1 - x^2}}$$

$$1 - x^2 \geq 0$$

$$x^2 - 1 \leq 0$$

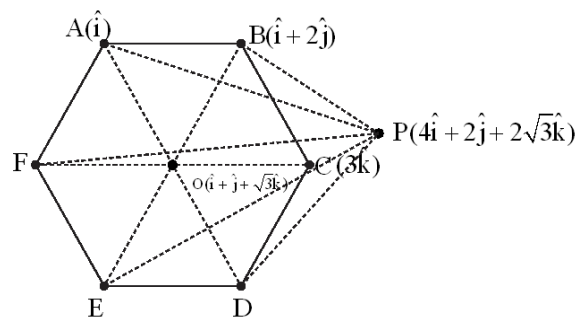
$$x \in [-1, 1] \quad \dots\dots D_2$$

$$\text{domain of } f(x) = D_1 \cap D_2 = [0, 1]$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (B,D)**



Equation of base plane, $x - 1 = 0$

$$\vec{OE} = -\vec{OB} \Rightarrow \vec{E} = 2\vec{O} - \vec{B} = \hat{i} + 2\sqrt{3}\hat{k}$$

$x = 1, y = 2, z = 2\sqrt{3}$ [Foot of perpendicular to base surface]

$$\text{Line AD; } \vec{r} = \hat{i} + \lambda(\hat{j} + \sqrt{3}\hat{k})$$

and $h = 3$.

6. **Ans (C,D)**

$$A = \{4, -2, -3, 2, 0\}$$

$$B = \{6, -10, 1, 4, 2\}$$

(A) Number of element in $A \times B$ are 25

$$\text{Number of subsets} = 2^{25}$$

(B) Number of function from set A to set B are 5^5

(C) Number of injective functions from set A to set

$$B \text{ are } {}^5P_5 = 5! = 120$$

(D) Number of onto functions, $5^5 - {}^5C_1 \cdot 4^5 + {}^5C_2 \cdot$

$$3^5 - {}^5C_3 \cdot 2^5 + {}^5C_4 \cdot 1^5 = 120.$$

7. Ans (A,B,C)

Let a point A on P_1 is (0,0,3)

Image of A about P_2 is $\left(\frac{24}{7}, \frac{-8}{7}, \frac{5}{7}\right)$;

Plane passing through common line of P_1 & P_2 is

$$P_1 + \lambda P_2 = 0$$

$$(2+3\lambda)x + (1-\lambda)y + (1-2\lambda)z + (-3-2\lambda) = 0$$

For image curve above plane will pass through

$$\left(\frac{24}{7}, \frac{-8}{7}, \frac{5}{7}\right)$$

$$\therefore \lambda = \frac{-3}{7} \text{ and plane equation : } 5x + 10y + 13z = 15$$

$$(B) \vec{n} \cdot \vec{n}_1 = 0 \left[\vec{n} = (2+3\lambda)\hat{i} + (1-\lambda)\hat{j} + (1-2\lambda)\hat{k} \right]$$

$$\therefore \lambda = -2 \text{ and plane equation : } 4x - 3y - 5z = 1$$

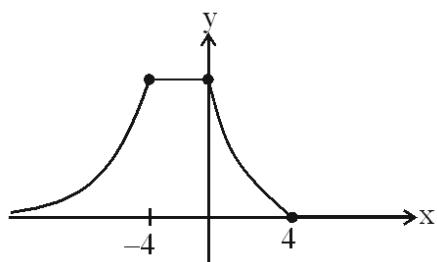
(C) Solving P_1 and P_2 ; let $z = \lambda$

$$\therefore x = 1 + \frac{\lambda}{5} \text{ \& } y = 1 - \frac{7\lambda}{5}$$

$$\Rightarrow \frac{x-1}{\frac{1}{5}} = \frac{y-1}{\frac{-7}{5}} = \frac{z-0}{1}$$

$$\text{Shortest distance} = \left| (\vec{a} - \vec{c}) \cdot \frac{(\vec{b} \times \vec{d})}{|\vec{b} \times \vec{d}|} \right| = \frac{5}{\sqrt{114}}$$

8. Ans (A,B,D)



$$\text{Let } x = 4 \tan \theta; \theta = \tan^{-1} \left(\frac{x}{4} \right); -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

$$f(x) = \begin{cases} 2\pi + 4\tan^{-1}(x/4) & x < -4 \\ \pi & -4 \leq x < 0 \\ \pi - 4\tan^{-1}(x/4) & 0 \leq x < 4 \\ 0 & 4 \leq x \end{cases}$$

9. Ans (B,C,D)

$$|\vec{b}| = 1, |\vec{c}| = 1, \vec{b} \cdot \vec{c} = 0$$

$\Rightarrow \vec{b}$ & $\vec{c}$ are perpendicular and $\vec{b} \times \vec{c}$ be a vector which is $\perp$ to $\vec{b}$ and $\vec{c}$ both.

$\Rightarrow$ Any vectors $\vec{a}$ can be written as

$$\vec{a} = x\vec{b} + y\vec{c} + z(\vec{b} \times \vec{c}) \quad \dots(i)$$

Now dot with $\vec{b}, \vec{c}, \vec{b} \times \vec{c}$

$$\vec{a} \cdot \vec{b} = x|\vec{b}|^2 + 0 + 0 \Rightarrow x = \vec{a} \cdot \vec{b}$$

$$\vec{a} \cdot \vec{c} = 0 + y + 0 \Rightarrow y = \vec{a} \cdot \vec{c}$$

$$(\vec{b} \times \vec{c}) \cdot \vec{a} = 0 + 0 + z|\vec{b} \times \vec{c}|^2$$

$$\Rightarrow z = \frac{\vec{a} \cdot (\vec{b} \times \vec{c})}{|\vec{b} \times \vec{c}|^2} = [\vec{a} \vec{b} \vec{c}]$$

From equation (i)

$$\Rightarrow \vec{a} = (\vec{a} \cdot \vec{b})\vec{b} + (\vec{a} \cdot \vec{c})\vec{c} + [\vec{a} \vec{b} \vec{c}](\vec{b} \times \vec{c})$$

$$\text{Now, } \vec{a} + \vec{b} = (\vec{a} \cdot \vec{b} + 1)\vec{b} + (\vec{a} \cdot \vec{c})\vec{c} + [\vec{a} \vec{b} \vec{c}](\vec{b} \times \vec{c})$$

$$\vec{a} + \vec{c} = (\vec{a} \cdot \vec{b})\vec{b} + (\vec{a} \cdot \vec{c} + 1)\vec{c} + [\vec{a} \vec{b} \vec{c}](\vec{b} \times \vec{c})$$

10. Ans (A,D)

3 digit no of the form $9k + 2$ are

$$\{101, 110, \dots, 992\}$$

Sum of 3 digit no of the form $(9k+2)$

$$= \frac{100}{2}(1093)$$

Similarly, sum of 3 digit no of the form $(9k+4)$

$$= \frac{100}{2}(1097)$$

$$\therefore \frac{100}{2}(1093) + \frac{100}{2}(1097)$$

$$= 100 \times 1095 = 500 \times 219$$

$$\therefore \lambda = 5.$$

11. Ans (A,B,D)

$$\alpha = \sum_{k=1}^{\infty} \tan^{-1} \left(\sqrt{k+2} \right) - \tan^{-1} \left(\sqrt{k} \right)$$

$$= \frac{3\pi}{4} - \tan^{-1} \sqrt{2}$$

$$\sin \alpha = \frac{\sqrt{2}+1}{\sqrt{6}}; \sin 2\alpha = \frac{1}{3}; \sin 4\alpha = -\frac{4\sqrt{2}}{9}$$

$$\tan 2\alpha = \frac{-1}{2\sqrt{2}}, \cos 2\alpha = -\frac{2\sqrt{2}}{3}$$

12. Ans (A,B,C,D)

$$f(x) = \begin{cases} 3x & x \geq 0 \\ x & x < 0 \end{cases}$$

$$g(x) = \begin{cases} \frac{x}{3} & x \geq 0 \\ x & x < 0 \end{cases}$$

$f(x)$ & $g(x)$ are inverse of each other

$$\therefore h(x) = x$$

$$k(x) = \cos^{-1}x + \cot^{-1}x; x \in [-1, 1]$$

$k(x)$ is decreasing function

$$K_{\max} = k(-1) = \pi + \frac{3\pi}{4} = \frac{7\pi}{4}$$

$$K_{\min} = k(1) = 0 + \frac{\pi}{4} = \frac{\pi}{4}$$

$$f(x) - g(x) = \begin{cases} \frac{8x}{3} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$\text{Domain; } \left(-\infty, \frac{3}{8}\right]$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (6.00)

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ 0 & 0 & 1 \end{vmatrix} - \begin{vmatrix} x_3 & y_3 & z_3 \\ x_4 & y_4 & z_4 \\ 0 & 0 & 1 \end{vmatrix} =$$

$$\begin{vmatrix} x_1x_3 + y_1y_3 + z_1z_3 & x_1x_4 + y_1y_4 + z_1z_4 & z_1 \\ x_2x_3 + y_2y_3 + z_2z_3 & x_2x_4 + y_2y_4 + z_2z_4 & z_2 \\ z_3 & z_4 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} \vec{a}_1 \cdot \vec{a}_3 & \vec{a}_2 \cdot \vec{a}_3 \\ \vec{a}_1 \cdot \vec{a}_4 & \vec{a}_2 \cdot \vec{a}_4 \end{vmatrix}$$

$$= (\vec{a}_1 \cdot \vec{a}_3)(\vec{a}_2 \cdot \vec{a}_4) - (\vec{a}_2 \cdot \vec{a}_3)(\vec{a}_1 \cdot \vec{a}_4)$$

$$= (\vec{a}_1 \times \vec{a}_2) \cdot (\vec{a}_3 \times \vec{a}_4)$$

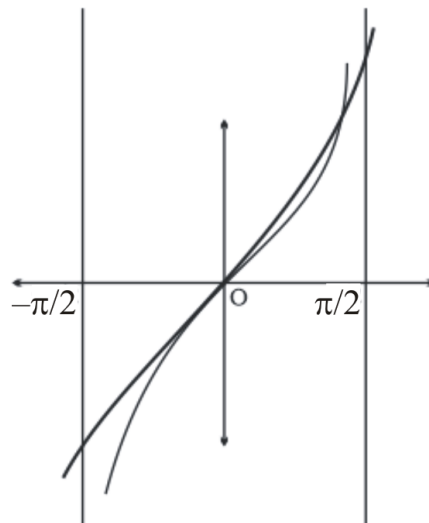
$$= |\vec{a}_1 \times \vec{a}_2| |\vec{a}_3 \times \vec{a}_4| \cos \frac{\pi}{3} = 6$$

2. Ans (2.00)

$f(x)$ and $g(x)$ both are increasing $\Rightarrow g(f(x))$ is increasing $\Rightarrow g(f(x)) = x$

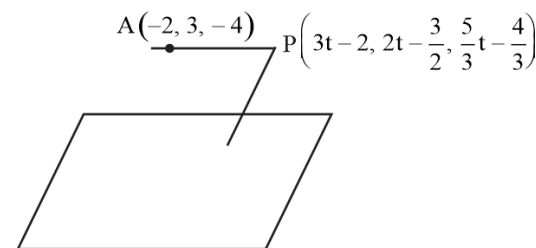
$$\Rightarrow \tan^{-1} \left(x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5} \right) = x$$

$$\Rightarrow x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5} = \tan x$$



$\Rightarrow 2$ solutions

3. Ans (8.50)



$$\text{Let } P \left(3t - 2, 2t - \frac{3}{2}, \frac{5}{3}t - \frac{4}{3} \right)$$

$$\text{Now dir of AP is } \left(3t, 2t - \frac{9}{2}, \frac{5}{3}t + \frac{8}{3} \right)$$

Now AP is parallel to the plane

$$\Rightarrow 4(3t) + 12 \left(2t - \frac{9}{2} \right) - 3 \left(\frac{5}{3}t + \frac{8}{3} \right) = 0$$

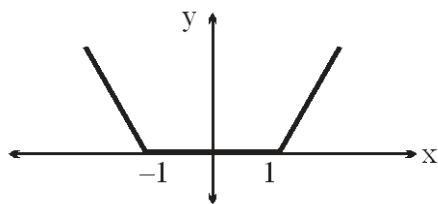
$$\Rightarrow 12t + 24t - 54 - 5t - 8 = 0 \Rightarrow t = 2$$

$$\Rightarrow \text{then point } P \left(4, \frac{5}{2}, 2 \right) \text{ and distance}$$

$$= \sqrt{(4+2)^2 + \left(3 - \frac{5}{2} \right)^2 + (-4-2)^2}$$

$$= \sqrt{36 + 36 + \frac{1}{4}} = \frac{17}{2} = 8.50$$

4. Ans (3.00)



$$f(x) = |2x - |x + 1|| + |x - 1||$$

Range of f is $[0, \infty)$

$$g(x) = \frac{1-x^2}{1+x^2} = \frac{2}{1+x^2} - 1$$

Range of $g(x)$ is $(-1, 1]$

$$\therefore \text{range of } f(g(x)) = \{0\} \Rightarrow \ell = 1$$

$$\text{range of } g(f(x)) = (-1, 1] \Rightarrow m = 2$$

$$\Rightarrow \ell + m = 3$$

5. Ans (442.50)

$$T_r = \tan^{-1} \left(\frac{32r}{16r^4 - 8r^2 + 17} \right)$$

$$= \tan^{-1} \left(\frac{2r}{r^4 - \frac{1}{2}r^2 + \frac{17}{16}} \right)$$

$$= \tan^{-1} \left(\frac{2r}{1 + r^4 - \frac{1}{2}r^2 + \frac{1}{16}} \right)$$

$$= \tan^{-1} \left(\frac{2r}{1 + (r^2 - \frac{1}{4})^2} \right)$$

$$= \tan^{-1} \left(\frac{2r}{1 + (r + \frac{1}{2})^2 (r - \frac{1}{2})^2} \right)$$

$$= \tan^{-1} \left(\frac{(r + \frac{1}{2})^2 - (r - \frac{1}{2})^2}{1 + (r + \frac{1}{2})^2 \cdot (r - \frac{1}{2})^2} \right)$$

$$T_r = \tan^{-1} \left(r + \frac{1}{2} \right)^2 - \tan^{-1} \left(r - \frac{1}{2} \right)^2$$

Put $r = 1, 2, 3, \dots, n$

$$T_1 = \tan^{-1} \left(\frac{3}{2} \right)^2 - \tan^{-1} \left(\frac{1}{2} \right)^2$$

$$T_2 = \tan^{-1} \left(\frac{5}{2} \right)^2 - \tan^{-1} \left(\frac{3}{2} \right)^2$$

$$T_n = \tan^{-1} \left(n + \frac{1}{2} \right)^2 - \tan^{-1} \left(n - \frac{1}{2} \right)^2$$

$$S_n = \tan^{-1} \left(n + \frac{1}{2} \right)^2 - \tan^{-1} \left(\frac{1}{2} \right)$$

$$S_n + \cot^{-1} 4 = \tan^{-1} \left(n + \frac{1}{2} \right)^2$$

$$\tan(S_n + \cot^{-1} 4) = \left(n + \frac{1}{2} \right)^2$$

$$= n^2 + n + \frac{1}{4}$$

$$\sum_{n=1}^{10} \tan(S_n + \cot^{-1} 4)$$

$$= \sum_{n=1}^{10} n^2 + n + \frac{1}{4}$$

$$= \frac{(10)(11)(21)}{6} + \frac{(10)(11)}{2} + \frac{1}{4} \times 10$$

$$= 442.50$$

6. Ans (285.00)

Lines are

$$L_1 = \frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1} = \lambda(\text{say})$$

$$L_2 = \frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4} = \mu(\text{say})$$

Co-ordinate of points $P(3\lambda+3, -\lambda+8, \lambda+3)$ and $Q(-3\mu-3, 2\mu-7, 4\mu+6)$ on line L_1 and L_2 .

If PQ is line of shortest distance, then direction ratios of

PQ is $3\lambda+3\mu+6, -\lambda-2\mu+15, \lambda-4\mu-3$, now PQ is perpendicular to both line L_1 & L_2 .

$$\Rightarrow 3(3\lambda+3\mu+6) - 1(-\lambda-2\mu+15) + 1$$

$$(\lambda-4\mu-3) = 0 \Rightarrow (11\lambda+7\mu=0)$$

$$\text{and } 3(3\lambda+3\mu+6) + 2(-\lambda-2\mu+15) + 4$$

$$(\lambda-4\mu-3) = 0 \Rightarrow 7\lambda+29\mu=0$$

$$\Rightarrow \lambda = \mu = 0$$

$$\Rightarrow P(3, 8, 3), Q(-3, -7, 6)$$

$$\Rightarrow \text{Length} = 3\sqrt{30} \text{ and}$$

Equation of line of shortest distance be

$$\frac{x-3}{6} = \frac{y-8}{15} = \frac{z-3}{-3}$$

$$\frac{x-3}{2} = \frac{y-8}{5} = \frac{z-3}{-1}$$

$$\Rightarrow a=2, b=5, d=3\sqrt{30}$$

$$\Rightarrow (a+b+c+d^2) = 2+5+8+270 = 285.00$$

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C,D	B,C,D	B,C	A	C	B,D	A,B,C	A,B,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	A	A	C	D				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	3.00	3.50 to 3.95	17.14 to 17.15	18.75	6.00	20.00		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A	B,C,D	C,D	B,C,D	A,D	A,B,C,D	A,B,C,D	A,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	B	D	A	C				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	8.00	7.00	8.00	4.00	7.00	0.00		

PART-3 : MATHEMATICS

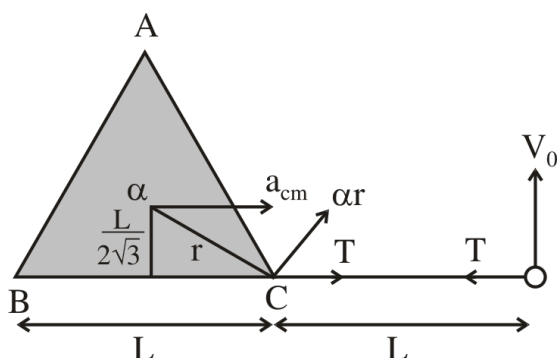
SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,D	A,B,C,D	A,B,C	C,D	A,B,C	A,D	A,B,C	B,C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	B	A	C	D				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	17.00	0.33	0.25	294.00	2.71 to 2.72	22.50		

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

2. Ans (B,C,D)



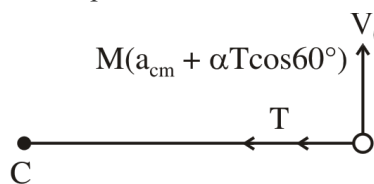
For plate: $T = M a_{cm}$

$$T \frac{L}{2\sqrt{3}} = I_{cm} \alpha = \frac{ML^2}{24} \alpha$$

$$\Rightarrow T = \frac{ML\alpha}{4\sqrt{3}} \Rightarrow Ma_{cm} = \frac{ML\alpha}{4\sqrt{3}}$$

$$\therefore a_{\text{cm}} = \frac{L\alpha}{4\sqrt{3}}$$

Path of particle will be circular with respect to point C.



Acceleration of C towards right is $(a_{cm} + \alpha r \cos 60^\circ)$.

With respect to C :

$$\mathbf{T} + \mathbf{M} \left(\mathbf{a}_{\text{cm}} + \alpha \frac{\mathbf{L}}{\sqrt{3}} \times \frac{1}{2} \right) = \frac{M \mathbf{v}_0^2}{L}$$

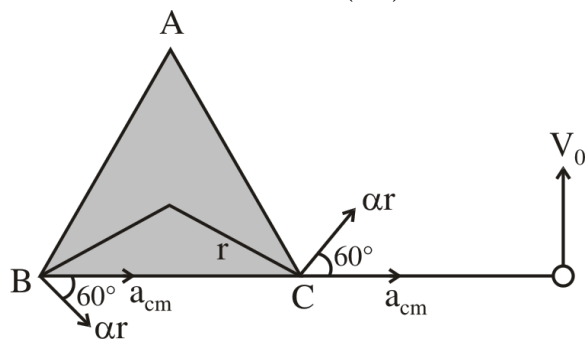
$$\Rightarrow T + Ma_{cm} + \frac{M\alpha L}{2\sqrt{3}} = \frac{Mv_0^2}{L}$$

$$\Rightarrow T + T + 2T = \frac{Mv_0^2}{L}$$

$$\Rightarrow 4T = \frac{Mv_0^2}{L} \Rightarrow T = \frac{Mv_0^2}{4L}$$

Radius of curvature of path of point mass M with respect to ground just after projection

$$= R = \frac{v_0^2}{a_{\perp}} = \frac{v_0^2}{(T/M)} = \frac{v_0^2}{\left(\frac{v_0^2}{4L}\right)} = 4L$$



Clearly acceleration of B and C will be same in magnitude just after the projection.

3. Ans (B,C)

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{(-20)} = \frac{1}{(-25)}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{20} - \frac{1}{25} \Rightarrow \frac{1}{v} = \frac{5-4}{100} \Rightarrow v = 100$$

$$\therefore m = \frac{-v}{u} \Rightarrow m = \frac{-100}{-20} = 5$$

$\therefore$ Radius of image circle = 5 mm

$$\text{Acceleration} = \sqrt{(\omega^2 R)^2 + (\alpha R)^2} = R\sqrt{\omega^4 + \alpha^2}$$

$\therefore$ Acceleration $\propto R$ as ω and α will be same for object and image.

$$\therefore \frac{\text{acceleration of image}}{\text{acceleration of object}} = \frac{5}{1} = 5$$

$$\text{Velocity of object} = \omega R = \alpha t \cdot R = 1 \times 2 \times 1 = 2 \text{ mm/s}$$

$$\text{Velocity of image} = \alpha \cdot t \cdot R' = 1 \times 2 \times 5 = 10 \text{ mm/s}$$

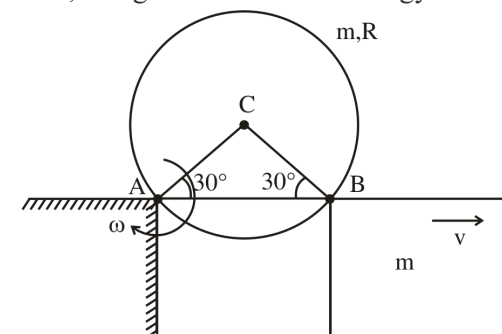
$\therefore$ Relative velocity = $10 - 2 = 8 \text{ mm/s}$ as both velocities are in same direction.

4. Ans (A)

$$v \cos 30^\circ = \omega R \sqrt{3} \cos 60^\circ$$

$$v = \omega R$$

Now, using conservation of energy



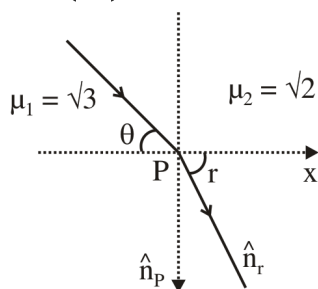
$$mgR(1 - \sin 30^\circ) = \frac{1}{2} \times 2mR^2\omega^2 + \frac{1}{2}mv^2$$

$$\frac{mgR}{2} = \frac{3}{2}mv^2$$

$$\Rightarrow gR = 3v^2$$

$$\Rightarrow v = \sqrt{\frac{gR}{3}} = \sqrt{\frac{10 \times 0.3}{3}} = 1 \text{ m/s}$$

5. Ans (C)



Angle between normal x-axis and incident ray is

$$(5\sqrt{2}\hat{i} - 5\hat{j} - 5\hat{k}) \cdot (\hat{i}) = 10 \cdot 1 \cdot \cos \theta$$

$$\cos \theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$$

Applying Snell's Law at P

$$\mu_1 \sin \theta = \mu_2 \sin r$$

$$\sqrt{3} \cdot \frac{1}{\sqrt{2}} = \sqrt{2} \sin r; r = 60^\circ$$

The unit vector along the refracted ray

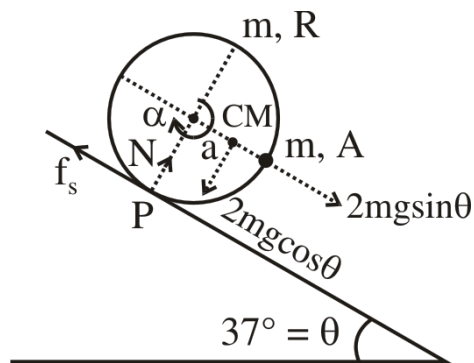
$$\hat{n}_r = \cos r \hat{i} + \sin r \hat{n}_p$$

$$\hat{n}_r = \cos 60^\circ \hat{i} + \sin 60^\circ \left(\frac{-\hat{j} - \hat{k}}{\sqrt{2}} \right)$$

$$\hat{n}_r = \frac{1}{2} \hat{i} + \frac{\sqrt{3}}{2\sqrt{2}} (-\hat{j} - \hat{k})$$

$$\hat{n}_r = \frac{\sqrt{2}\hat{i} - \sqrt{3}\hat{j} - \sqrt{3}\hat{k}}{\sqrt{8}}$$

6. Ans (B,D)



For pure rolling motion, $a = \alpha R$

$$2mg \cos \theta \times \frac{R}{2} + 2mg \sin \theta \times R = 4mR^2 \alpha$$

$$mg(\cos \theta + 2 \sin \theta) = 4m\alpha R$$

$$g(\cos 37^\circ + 2 \sin 37^\circ) = 4\alpha R$$

$$g \left(\frac{4}{5} + 2 \times \frac{3}{5} \right) = 4\alpha R$$

$$2g = 4\alpha R \Rightarrow \alpha = \frac{g}{2R}$$

$$\therefore a = \alpha R = g/2$$

$$\text{Now, } 2mg \sin \theta - f_s = 2ma$$

$$\frac{6mg}{5} - f_s = 2m \frac{g}{2}$$

$$\Rightarrow f_s = \frac{6mg}{5} - mg = \frac{mg}{5} = 2N$$

$$\text{Also, } 2mg \cos \theta - N = 2m\alpha \frac{R}{2}$$

$$\frac{8mg}{5} - N = \frac{mg}{2}$$

$$\Rightarrow N = \frac{8mg}{5} - \frac{mg}{2} = \frac{11}{10}mg$$

$$N = 11N$$

8. Ans (A,B,D)

$$I_x = \frac{m(2\ell)^2}{12} + \frac{m\left(\frac{\ell}{4}\right)^2}{12} = \frac{65}{192}m\ell^2$$

$$I_y = \frac{m(2\ell)^2}{12} + m\left(\frac{\ell}{2}\right)^2 + \frac{m\left(\frac{\ell}{4}\right)^2}{12} + m\left(\frac{\ell}{2}\right)^2 = \frac{161}{192}m\ell^2$$

$$I_z = \frac{m(2\ell)^2}{6} + m\left(\frac{\ell}{2}\right)^2 + m\left(\frac{\ell}{2}\right)^2 = \frac{7}{6}m\ell^2$$

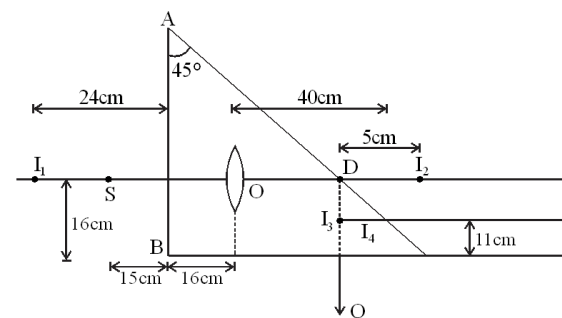
PART-1 : PHYSICS

SECTION-I (ii)

10. Ans (A)

Image is formed at I_1 at a distance $15 \times 1.6 = 24$ cm from AB

Refraction at lens



Focal length of the lens.

$$\frac{1}{f} = \left(\frac{2.4}{1.6} - 1 \right) \left(\frac{2}{20} \right) \Rightarrow f = +20 \text{ cm}$$

Object distance = $24 + 16 = 40$ cm

$$\text{Lens formula } \frac{1}{v} - \frac{1}{-40} = \frac{1}{+20}$$

$$v = +40 \text{ cm}$$

Image is at I_2

OD = 35 cm (from geometry of the figure)

$$DI_2 = 5 \text{ cm}$$

$\therefore$ Light ray after passing through the lens will suffer total internal reflection at face AC. This is because angle of incidence is close to 45° , as we are considering paraxial rays.

$$45^\circ > \sin^{-1} \left(\frac{1}{1.6} \right)$$

Surface AC may be treated as a mirror for which object is at I_2 .

Image is formed at I_3 .

$$DI_3 = 5 \text{ cm}$$

Observer at P will not be able to see image as light rays suffer TIR at surface AC. Observer at Q

see the image at I_4 image I_4 is above surface

$$BC \text{ by a distance} = \frac{11}{1.6} = \frac{55}{8} \text{ cm}$$

11. Ans (C)



Let u be the speed of A just before come in contact with the spring.

$$\therefore mgh = \frac{1}{2}mu^2 + \frac{1}{2}I\omega^2 = \frac{mu^2}{2} + \frac{I}{2} \frac{u^2}{r^2}$$

$$\Rightarrow mgh = \frac{u^2}{2} \left(m + \frac{I}{r^2} \right) \Rightarrow mgh = \frac{u^2}{2} \left(\frac{mr^2 + I}{r^2} \right)$$

$$\Rightarrow u = \sqrt{\frac{2mgh r^2}{mr^2 + I}}$$

At maximum compression both have same linear speed and as both are rolling purely being identical both have same angular speed too. Let common velocity at maximum compression be v' .

From momentum conservation $v' = \frac{u}{2}$

$$= \sqrt{\frac{mgh r^2}{(mr^2 + I) 2}} \quad \{\text{Note that net friction force on system} = 0\}$$

$$\omega' = \frac{v'}{r} = \frac{u}{2r}$$

Since work done by friction is zero here, mechanical energy is conserved.

From start of compression to maximum compression :

$$\frac{1}{2}mu^2 + \frac{1}{2}I\omega^2 = 2 \times \left[\frac{1}{2}mv'^2 + \frac{1}{2}I\omega'^2 \right] + \frac{1}{2}kx^2$$

$$= m \left(\frac{u}{2} \right)^2 + I \left(\frac{\omega}{2} \right)^2 + \frac{1}{2}kx^2$$

$$\Rightarrow \frac{1}{2} \left[\frac{mu^2}{2} + \frac{I\omega^2}{2} \right] + \frac{kx^2}{2}$$

$$\Rightarrow \frac{1}{2}mgh = \frac{kx^2}{2} \Rightarrow x = \sqrt{\frac{mgh}{k}}$$

$\therefore$ Maximum compression is same for all objects. Ratio of maximum compression in spring to maximum

$$\text{velocity of object A} = \frac{\sqrt{\frac{mgh}{k}}}{\sqrt{\frac{2mgh r^2}{mr^2 + I}}} = \sqrt{\frac{mgh}{k} \times \frac{mr^2 + I}{2mgh r^2}}$$

$$= \sqrt{\frac{(mr^2 + I)}{2k r^2}}$$

$$\therefore \text{For solid sphere, ratio} = \sqrt{\frac{mr^2 + \frac{2}{5}mr^2}{2k r^2}} = \sqrt{\frac{7m}{10k}}$$

$$\text{Hollow sphere} \rightarrow \sqrt{\frac{mr^2 + \frac{2}{3}mr^2}{2k r^2}} = \sqrt{\frac{5m}{6k}}$$

$$\text{Ring} \rightarrow \sqrt{\frac{mr^2 + mr^2}{2k r^2}} = \sqrt{\frac{m}{k}}$$

$$\text{Disc} \rightarrow \sqrt{\frac{mr^2 + \frac{mr^2}{2}}{2k r^2}} = \sqrt{\frac{3m}{4k}}$$

12. Ans (D)

Maximum speed of A is just before touching the spring.

$$u = \sqrt{\frac{2mgh r^2}{mr^2 + I}}$$

$$\text{Fore maximum compression } mgh = \frac{1}{2}kx^2$$

$$x = \sqrt{\frac{2mgh}{k}}$$

$$\therefore \text{Ratio} = \frac{\sqrt{\frac{2mgh}{k}}}{\sqrt{2} \cdot u} = \sqrt{\frac{(mr^2 + I)}{2kr^2}}$$

Putting values we get the respective answers.

PART-1 : PHYSICS

SECTION-II

1. Ans (3.00)

When distance of object from lens is x , let the distance of image be y .

Lens formula

$$\frac{1}{-y} - \frac{1}{-x} = \frac{1}{-f}$$

$$\frac{1}{y} = \left(\frac{1}{x} + \frac{1}{f} \right) \Rightarrow y = \frac{xf}{x+f}$$

$$\text{Magnification } m = \frac{y}{x} = \frac{f}{x+f}$$

Height of image is $h_i = mh_0$ (h_0 = height of object)

$$h_i = \frac{f h_0}{x+f}$$

Angle subtended by the image at the eye

$$\theta = \frac{h_i}{D-x+y} = \frac{f h_0}{(x+f)(D-x+y)}$$

$$\text{Let } (x+f)(D-x+y) = Z$$

$$Z = \left(D-x + \frac{xf}{x+f} \right) (x+f)$$

$$Z = -x^2 + Dx + Df$$

'Z' is maximum when

$$\frac{dZ}{dx} = 0$$

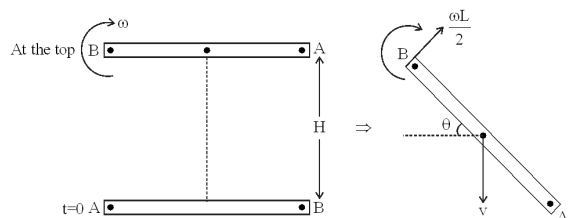
$$x = \frac{D}{2}$$

2. Ans (3.50 to 3.95)

$$\text{Time of flight } T = \frac{2v}{g} = \frac{2\pi\sqrt{gL}}{g} = 2\pi\sqrt{\frac{L}{g}}$$

$$\therefore \omega = \frac{2\pi}{T} = \sqrt{\frac{g}{L}}$$

As per problem rod rotates by 180° by time it's center is at top. But after this time also end B continues to rise in its vertical direction while the centre of rod would be coming down. Let θ = angle made by rod with horizontal at the instant when vB is horizontal. t be time spent after the centre of rod was at its topmost position.



$$\Rightarrow \frac{\omega L \cos \theta}{2} = gt$$

$$\Rightarrow \sqrt{\frac{g}{L}} \cdot \frac{L}{2} \cdot \cos \left(\sqrt{\frac{g}{L}} \cdot t \right) = gt$$

$$\Rightarrow \cos \left(\sqrt{\frac{g}{L}} \cdot t \right) = 2t\sqrt{\frac{g}{L}}$$

As per question, solution of $\cos \theta = 2\theta$ is $\theta = 0.45$

$$\therefore \sqrt{\frac{g}{L}} \cdot t = 0.45$$

$$\Rightarrow t = 0.45 \sqrt{\frac{L}{g}} = 0.45 \text{ sec}$$

Time taken by centre of rod to reach top = $\frac{v}{g}$

$$= \frac{\pi\sqrt{gL}}{g} = \pi\sqrt{\frac{L}{g}} = \pi = 3.14 \text{ sec}$$

$\therefore$ Total time taken by point B to reach its topmost position = $0.45 + 3.14 = 3.59 \text{ sec}$.

3. Ans (17.14 to 17.15)

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} + \frac{1}{36} = \frac{1}{20} \Rightarrow \frac{1}{v} = \frac{9-5}{180}$$

$$\frac{1}{v} = \frac{4}{180} \Rightarrow 45 \text{ cm}$$

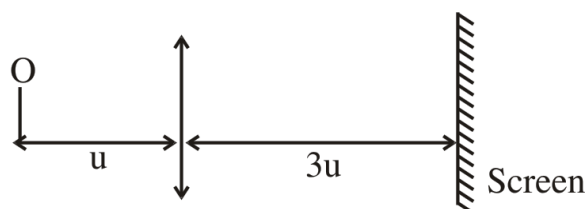
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}; \frac{1}{20} + \frac{1}{120} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{f} = \frac{6+1}{120}$$

$$\frac{1}{f} = (\mu - 1) \left[\frac{2}{R} \right] = \frac{1}{2} \Rightarrow \frac{7}{120} = \frac{1}{(4R)}$$

$$\Rightarrow R = \frac{120}{7} = 17.14 \text{ to } 17.15$$

4. Ans (18.75)



$$\frac{h_1}{h_0} = m \Rightarrow h_1 = mh_0$$

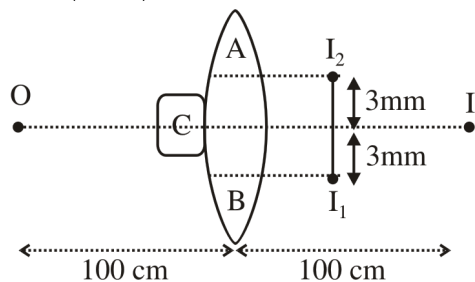
$$\frac{h_1}{h_0} = \frac{1}{m} \Rightarrow h_2 = \frac{h_0}{m}$$

$$\frac{h_1}{h_2} = \frac{m^2}{1} = 9 : 1 \Rightarrow m = 3$$

$$4u = 100 \Rightarrow u = 25 \text{ cm}$$

$$\frac{1}{75} + \frac{1}{25}$$

5. Ans (6.00)



The image I of point object O formed by the part C is at a distance 100 cm right on the optic axis of the lens part C.

Now,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{100} = \frac{1}{50}$$

$$v = +\frac{100}{3} \text{ cm}$$

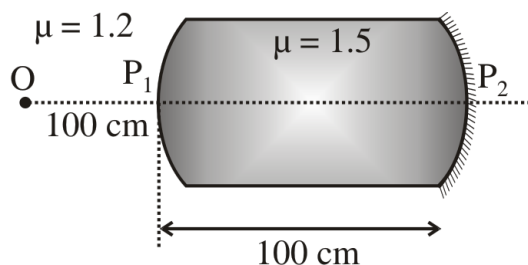
$$\therefore m = \frac{v}{u} = +\frac{1}{3}$$

$\therefore$ The distance between the two images formed by the parts A and B

$$d = 3 \text{ mm} + 3 \text{ mm} = 6 \text{ mm}$$

6. Ans (20.00)

Refraction from P_1



$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1.5}{v_1} - \frac{1.2}{-100} = \frac{1.5 - 1.2}{+20}$$

$$\Rightarrow v_1 = 500 \text{ cm}$$

Reflection from P_2

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{-200} = \frac{1}{v_2} + \frac{1}{400}, v_2 = -\frac{400}{3} \text{ cm}$$

Again refraction from P_1

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1.2}{v_3} - \frac{1.5}{100/3} = \frac{1.2 - 1.5}{-20}$$

$$\Rightarrow v_3 = +20 \text{ cm (left to } P_1)$$

PART-2 : CHEMISTRY

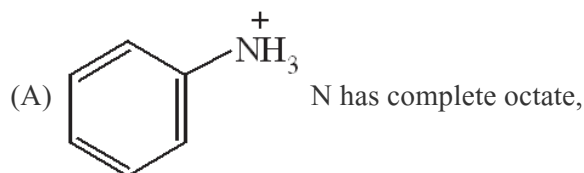
SECTION-I (i)

1. Ans (A)

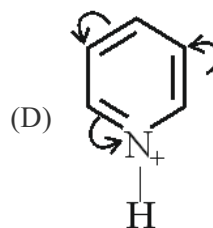
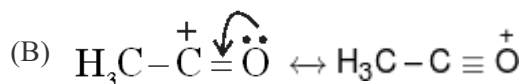
Na/liq NH_3 reduces alkyne into trans double bond.

$\text{H}_2/\text{Pd}, \text{BaSO}_4$ reduces alkyne into C is double.

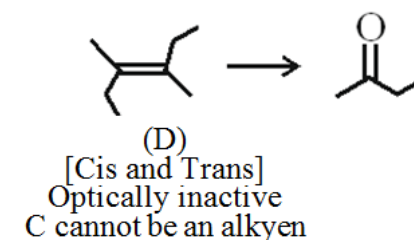
2. Ans (B,C,D)



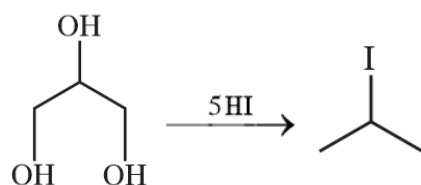
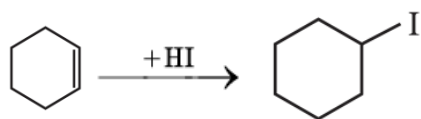
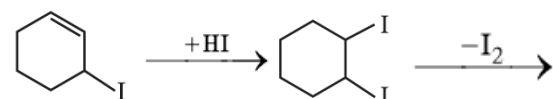
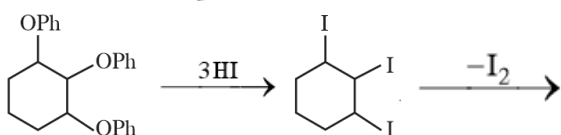
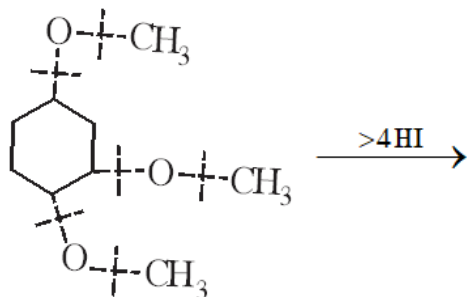
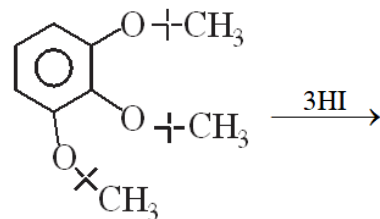
so can't accept electrons from benzene ring.



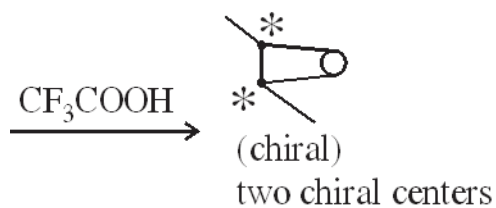
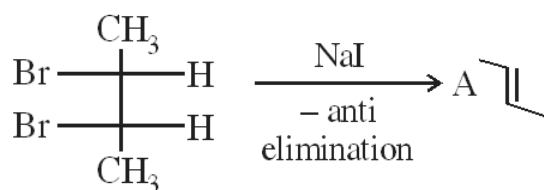
3. Ans (C,D)



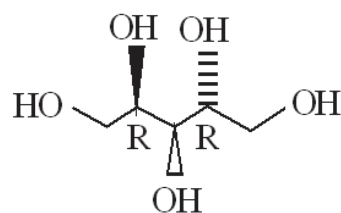
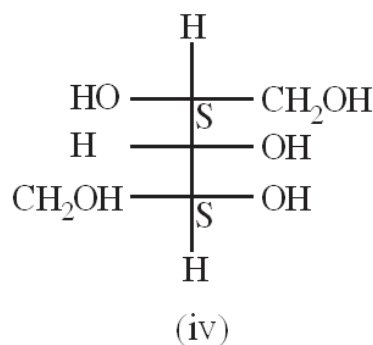
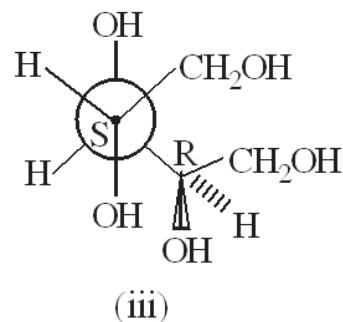
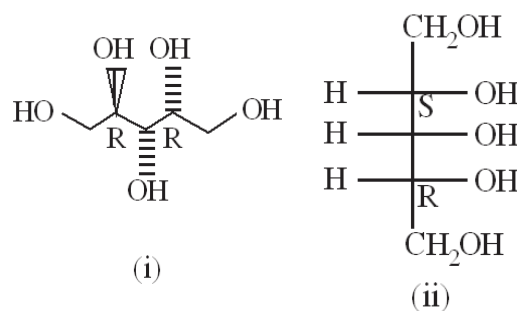
4. Ans (B,C,D)



5. Ans (A,D)



6. Ans (A,B,C,D)



i, v $\rightarrow$ identical

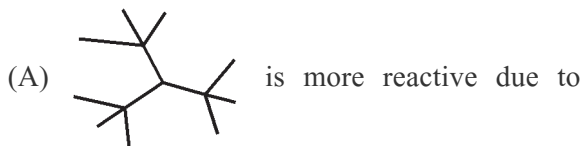
i, iv $\rightarrow$ enantiomers

ii, iii $\rightarrow$ diastereomers & meso compounds.

7. Ans (A,B,C,D)

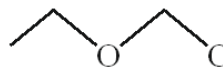
All are more acidic enough to decompose NaHCO_3 .

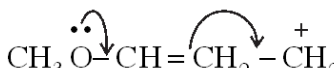
8. Ans (A,D)



large steric strain.

(B) H_2O is more ionising than ethanol, so S_N^1 reactions proceed faster in H_2O .

(C) in , oxygen stabilises the carbocation in S_N^1 & T.S in S_N^2 .

(D)  is more stable.

PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (B)

III is aromatic and also an alkene and have no active hydrogen

II is less acidic than EtOH

10. Ans (D)

I is insoluble in NaOH

IV is not aromatic and have no active H.

11. Ans (A)

ether & 1° OH fails in Lucas Test

ether has lower boiling points than water

ether do not have active H.

12. Ans (C)

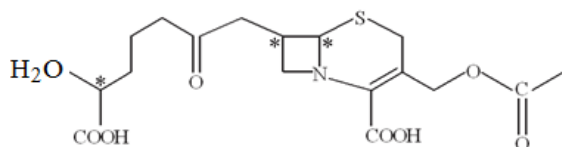
Boiling point of $1^\circ\text{OH} > \text{H}_2\text{O} \geq 2^\circ\text{OH} > 3^\circ\text{OH}$.

3°OH does not undergo oxidation.

PART-2 : CHEMISTRY

SECTION-II

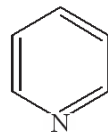
1. Ans (8.00)



3 chiral centers no. of stereoisomers = 8.

2. Ans (7.00)

$\cdot\dot{\text{C}}\text{H}_2$ triple is a radical with no vacant orbital



is nucleophile.

MnO_4^- with Mn^{+7} acts as an electrophile & adds to alkene.

3. Ans (8.00)

Can't exhibit geometrical isomerism as it is non planar.

4. Ans (4.00)

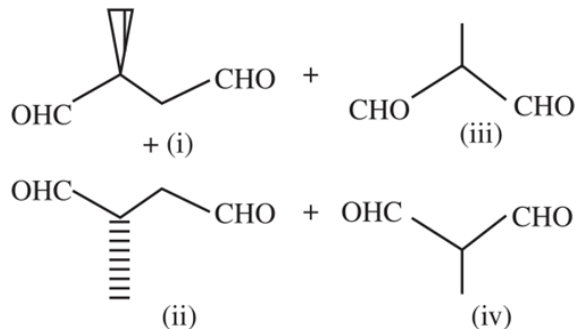
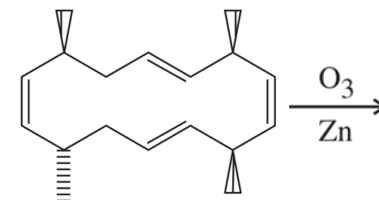
i, iii, iv, vii



5. Ans (7.00)

ii, iii, iv, v, vii, viii, ix

6. Ans (0.00)



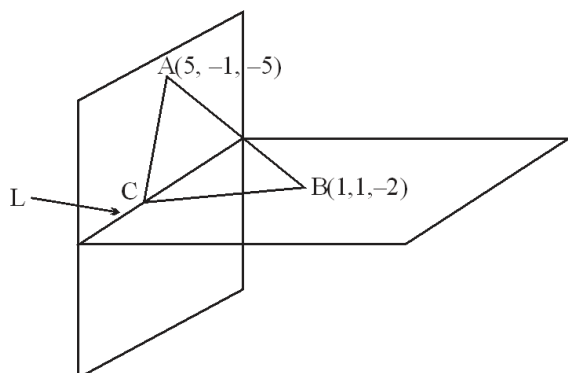
i & ii are identical

iii & iv are identical.

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,D)



$$AB = \sqrt{16 + 4 + 9} = \sqrt{29}$$

Shortest distance (SD) between L and AB:

where $\vec{AB} = 4\hat{i} - 2\hat{j} - 3\hat{k}$

vector along line L = $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & 0 \\ 3 & -1 & 1 \end{vmatrix} = 3\hat{i} - \hat{j} - 10\hat{k}$

vector along SD = $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -2 & -3 \\ 3 & -1 & -10 \end{vmatrix} = 17\hat{i} + 3\hat{j} + 2\hat{k}$

A point on L $\equiv (2, 0, -6) \equiv P$

$\vec{BR} = \hat{i} - \hat{j} - 4\hat{k}$

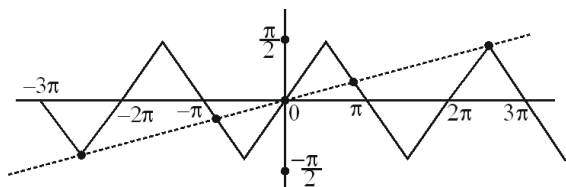
$\therefore SD = \left| \frac{17 - 31 - 8}{\sqrt{289 + 961 + 4}} \right| = \frac{22}{\sqrt{1254}} = \sqrt{\frac{22}{57}}$

and minimum area of $\Delta ABC = \sqrt{\frac{319}{114}}$

2. Ans (A,B,C,D)

$5 \sin^{-1}(\sin x) - x = 0$

$\sin^{-1}(\sin x) = x/5$



Verify the options.

3. Ans (A,B,C)

$\sin \theta (2 \sin \theta - 1) \geq 0 \Rightarrow \sin \theta \geq \frac{1}{2} \text{ or } \sin \theta \leq 0$

$A \equiv \left[\frac{\pi}{6}, \frac{5\pi}{6} \right] \cup [\pi, 2\pi] \cup \{0\}$

$\sin^{-1}(\cos \theta) \leq \cos^{-1}(\cos \theta)$

$\cos^{-1}(\cos \theta) \geq \frac{\pi}{4} \Rightarrow \cos \theta \leq \frac{1}{\sqrt{2}}$

$\Rightarrow B = \left[\frac{\pi}{4}, \frac{7\pi}{4} \right]$

$A \cap B = \left[\frac{\pi}{4}, \frac{5\pi}{6} \right] \cup \left[\pi, \frac{7\pi}{4} \right]$

$A \cup B = \{0\} \cup \left[\frac{\pi}{6}, 2\pi \right]$

4. Ans (C,D)

$|\vec{a}| = 1, |\vec{b}| = 2, |\vec{c}| = 3$

$|\vec{a} - 2\vec{b} + \vec{c}|^2 + |\vec{a} + \vec{b} - 2\vec{c}|^2 + |-2\vec{a} + \vec{b} + \vec{c}|^2 = 126,$

$\Rightarrow 6 (|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} - \vec{c} \cdot \vec{a}) = 126$

$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} - \vec{c} \cdot \vec{a} = 21$

$\Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -7$

$\Rightarrow |\vec{a} + \vec{b} + \vec{c}|^2 = \sum |\vec{a}|^2 + 2 \sum \vec{a} \cdot \vec{b} = 0$

$\Rightarrow \vec{a} + \vec{b} + \vec{c} = 0$

$\Rightarrow \vec{a} + \vec{b} = -\vec{c} \Rightarrow |\vec{a} + \vec{b}|^2 = |\vec{c}|^2 \Rightarrow \vec{a} \wedge \vec{b} = 0$

$||^{\wedge} \vec{b} \vec{c} = \vec{c} \wedge \vec{a} = \pi$

$\vec{b} = 2\vec{a}, \vec{c} = -3\vec{a}$

Now, verify the options.

5. Ans (A,B,C)

(A) & (B) obviously correct.

Number of symmetric relations on A such that

$n(A) = m$ is equal to $(2)^{\frac{m(m+1)}{2}} = 2^{10}.$

Number of reflexive relations on A such that n

$(A) = m$ is equal to $(2)^{m^2-m} = 2^{12}.$

Hence, options A,B,C are correct.

6. Ans (A,D)

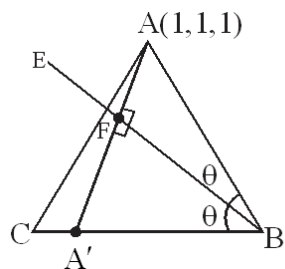


Image of A in BE lies on BC which is A' (5, 7, 9)

$\therefore$ equation of BC is $\frac{x-5}{4} = \frac{y-7}{3} = \frac{z-9}{3}$

Now point of intersection of BE and BC is (1, 4, 6).

7. Ans (A,B,C)

$x^x = x^x \Rightarrow U$ is reflexive

$x^y = y^x \Rightarrow y^x = x^y \Rightarrow U$ is symmetric

also U is transitive

(Because let $(x_1, y_1) \in U$ & $(y_1, z_1) \in U$, then (x_1, z_1) also belongs U)

8. Ans (B,C,D)

Put $x = y = 1$

$$f^2(1) + f^2(1) = 2f(1)$$

$$f^2(1) - 2f(1) + 1 = 0 \Rightarrow (f(1) - 1)^2 = 0.$$

$$\Rightarrow f(1) = 1$$

Now, $y = 1$ gives

$$f(x) = f\left(\frac{\lambda}{x}\right) \forall x > 0$$

$$\text{Now put } y = \frac{\lambda}{x}$$

$$f(x) f\left(\frac{\lambda}{x}\right) + f\left(\frac{\lambda}{x}\right) f(x) = 2f(\lambda)$$

$$\Rightarrow f(x) f\left(\frac{\lambda}{x}\right) = 1$$

$$\therefore f^2(x) = 1 \forall x > 0 \text{ [Since } f(x) = f(1/x)]$$

If we put $x = y = \sqrt{t}$

$$\Rightarrow f^2(\sqrt{t}) + f^2\left(\frac{\lambda}{\sqrt{t}}\right) = 2f(t) \Rightarrow f(x) = 1$$

$\Rightarrow$ So, $f(x)$ is periodic function, Range of $f(x)$

$$= \{1\}.$$

PART-3 : MATHEMATICS

SECTION-I (ii)

10. Ans (A)

$$|\vec{AB}| = \sqrt{1 + \alpha + \alpha^2} = |\vec{AC}| \text{ and } \vec{BC} = -2\sqrt{\alpha} \hat{j}$$

$$\Rightarrow |\vec{BC}| = 2\sqrt{\alpha}$$

$$\Rightarrow \vec{AB} \cdot \vec{AC} = 1 - \alpha + \alpha^2$$

$$\Rightarrow \cos A = \frac{1 - \alpha + \alpha^2}{1 + \alpha + \alpha^2} \leq 1$$

$$\Rightarrow 1 - \alpha + \alpha^2 \leq 1 + \alpha + \alpha^2 \Rightarrow \alpha \geq 0$$

$$\text{But } \frac{1 - \alpha + \alpha^2}{1 + \alpha + \alpha^2} \in \left[\frac{1}{3}, 1\right]$$

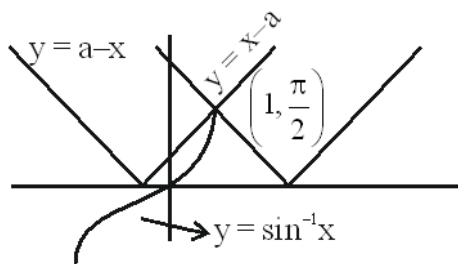
$$\therefore \cos A \geq \frac{1}{3}$$

$$\Rightarrow A_{\text{greatest}} \text{ for } \cos A = \frac{1}{3}$$

$$\Rightarrow A = \tan^{-1}(2\sqrt{2})$$

$$\Rightarrow \left|\frac{\Delta}{\alpha}\right| = \sqrt{\alpha + \frac{1}{\alpha}} \geq \sqrt{2}$$

12. Ans (D)



$$y = \sin^{-1} x$$

$y = x - a$ passes through

$$\left(1, \frac{\pi}{2}\right), \frac{\pi}{2} = 1 - a \Rightarrow a = 1 - \frac{\pi}{2}$$

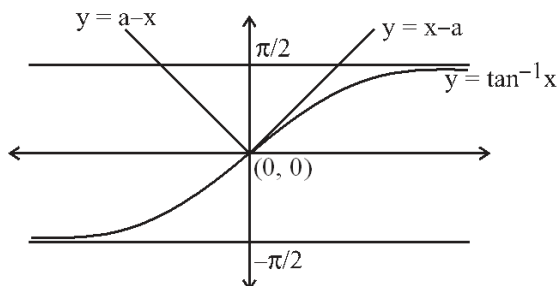
Also $y = a - x$ passes through $\left(1, \frac{\pi}{2}\right)$

$$\frac{\pi}{2} = a - 1 \Rightarrow a = 1 + \frac{\pi}{2}$$

$$\text{So } 1 - \frac{\pi}{2} \leq a \leq 1 + \frac{\pi}{2}$$

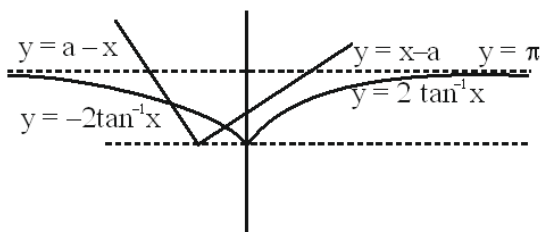
Integral values of 'a' is 0, 1, 2

(II) $y = \tan^{-1} x$



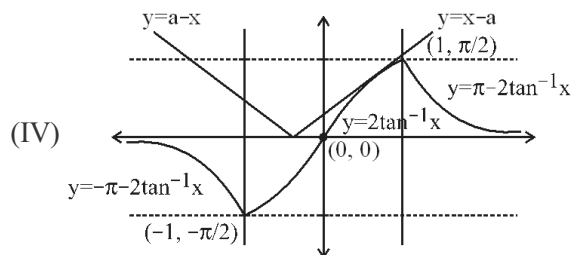
clearly $a \geq 0$

$$(III) y = \cos^{-1} \left(\frac{1 - x^2}{1 + x^2} \right)$$



Clearly the graph of $y = |x - a|$ cuts the graph of

$$y = \cos^{-1} \left(\frac{1 - x^2}{1 + x^2} \right) \text{ the all values of } a \in \mathbb{R}.$$



$$y = \sin^{-1} \left(\frac{2x}{1 + x^2} \right)$$

$y = x - a$ touches

$$y = 2 \tan^{-1} x$$

$$1 = \frac{2}{1 + x^2}$$

$$x = 1$$

$$y = 2 \tan^{-1}(1) = \frac{\pi}{2}$$

Point $\left(1, \frac{\pi}{2}\right)$ lies on $y = x - a$

$$\frac{\pi}{2} = 1 - a,$$

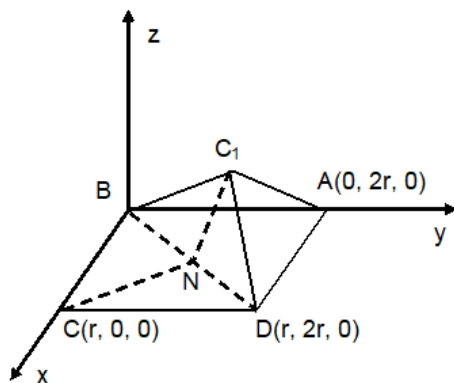
$$a \geq 1 - \frac{\pi}{2}, \text{ clearly integral values of } a = 0, 1, 2,$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (17.00)

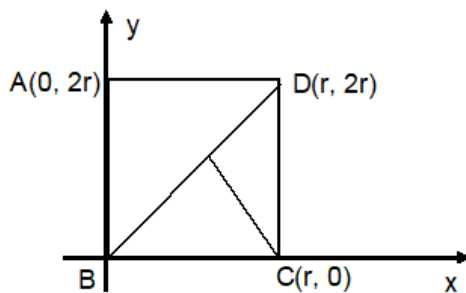
Let $r = \sqrt{85}$



Let the rectangle ABCD initially lies in xy plane with B lying at origin BC along x-axis and BA along y-axis. Equation of BD in xy plane is $y = 2x$. So, the coordinates of foot of the perpendicular N from C on BD are

$$(r/5, 2r/5) \text{ and length } CN = \frac{2r}{\sqrt{5}}$$

Clearly $CN = C_1N$

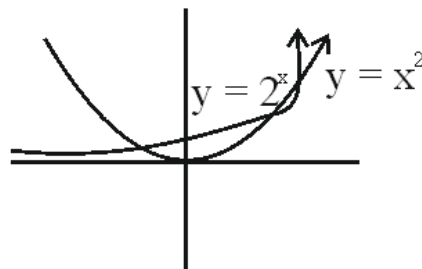


Hence the coordinates of various points in 3-D are
A (0, 2r, 0), C (r, 0, 0), D (r, 2r, 0), N (r/5, 2r/5, 0)
and $C_1(r/5, 2r/5, 2r/\sqrt{5})$

$$\text{Now } AC_1 = \frac{\sqrt{85} r}{5}$$

$$\Rightarrow AC_1 = 17$$

2. Ans (0.33)



Number of solution of equation $2^x = x^2$ is 3

Now $k = 3$

$$f(x) = 3x - [3x] + [|\sin(\cos x)|]$$

$$f(x) = \{3x\} + [|\sin(\cos x)|]$$

Since, $\{3x\}$ is periodic with period $1/3$ and

$$[|\sin(\cos x)|] = 0 \quad \forall x \in \mathbb{R}. \text{ So, answer is 0.33.}$$

3. Ans (0.25)

$$f(x)|_{\min.} = f\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi^3}{32} + 0 = \frac{\pi^3}{32}$$

$$f(x)|_{\max.} = f(-1) = \frac{7\pi^3}{8} + \frac{\pi^3}{8} = \pi^3$$

$$\therefore \text{Range} = \left[\frac{\pi^3}{32}, \pi^3\right]$$

$$\Rightarrow \frac{8a}{b} = \frac{1}{4} = 0.25.$$

4. Ans (294.00)

$$[\vec{a} \vec{b} \vec{c}]^2 = \begin{vmatrix} \sin^2 \alpha & 2 & -3 \\ 1 & 2\sin^2 \alpha & -2 \\ 1 & 1 & -\sin^2 \alpha \end{vmatrix}^2$$

$$= [2\sin^6 \alpha - 10\sin^2 \alpha + 7]^2$$

$$\text{Let } f(\alpha) = 2\sin^6 \alpha - 10\sin^2 \alpha + 7$$

$$\text{Let } \sin^2 \alpha = p : p \in [0, 1]$$

$$H(p) = 2p^3 - 10p + 7$$

$$H'(p) = 6p^2 - 10 < 0$$

$$\therefore f(x) \in [-1, 7] \Rightarrow [\vec{a} \vec{b} \vec{c}]^2 = [0, 49]$$

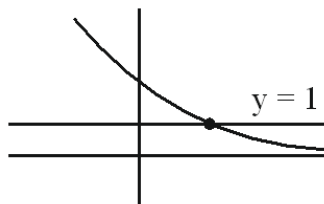
$$\therefore v = 6 [\vec{a} \vec{b} \vec{c}]^2 = [0, 294].$$

5. Ans (2.71 to 2.72)

$$3^x + 4^x + 5^x + \dots + n^x = (n+1)^x$$

divide by $(n+1)^x$

$$\left(\frac{3}{n+1}\right)^x + \left(\frac{4}{n+1}\right)^x + \left(\frac{5}{n+1}\right)^x + \dots + \left(\frac{n}{n+1}\right)^x = 1$$



clearly one value of 'x' satisfying the equation so $a = 1$.

Range of $y = \sin^{-1}(\sin^{-1}x)$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

So integers $-1, 0, 1$. So $f = 3$

$|ex^2 + e| = e|x^2 + 1|$ minimum value of expression is e .

6. Ans (22.50)

Placed student in IIT X = 261

Placed student in IIT Y = 272

Total placed by A + B + C = 280 + 300 + 185 = 765

$\therefore$ Placed student in IIT Z = 765 - 533 = 232

So number of unplaced students in

IIT'S = N = 29 + 68 + 128 = 225